Testing report and specification

ABC Fashion AI Shopping Agent

COMP3850 Deliverable 4 | Group 20 | Revised 24 October 2026

Prepared by Minh | Testing lead Sarah Thomas | Submission date 24 October 2026

Purpose and current position

This report specifies and records testing for the eight customer use cases in Figure 7 of the final D4 document (page 62), supported by the UC01–UC07 descriptions on pages 63–72 and the SRS on pages 38–53. It retains stable test identifiers, exact inputs, observed results and evidence links. Browse / Search Products is the eighth customer use case and is identified as BR in this report because the supplied descriptions leave it unnumbered.

On 24 October, all 48 offline scripts passed; 32 current catalogue/model/HTTP checks and all three isolated AI/database failure/recovery checks passed. The full 38-scenario real-model run produced 108 passes and six provider timeouts; three fresh repetitions of each affected scenario then passed (9/9). Deployed browser testing also found an intermittent occasion conflict, a search recovery defect (fixed locally) and a value-comparison constraint failure. These are retained in DF6–DF8; an all-passing technical readiness decision is not supported. [E14–E16]

Peyton Bernhardt accepted the product as ready at the 21 October meeting attended by all team members, as confirmed by Salvi. The demonstrated product decision is retained in E13. This report does not substitute that acceptance for technical evidence or Sarah Thomas's final review. The new SRS also contains administrator requirements outside the eight customer use cases; their execution is not established by the customer tests. [E13, E17]

Document control

The revision history records changes to the testing report (see TABLE 1).

TABLE 1 Report revision history

Version and date	Report contributor	Changes and reason
Source revision 19 October 2026	Original author not identified in supplied PDF	Original Part B functional cases, offline inventory, historical customer-scenario summary and outstanding tests.
1.1 22 October 2026	Minh	Added Part A and dated schedule; retained original IDs and historical results; clarified inputs; indexed 46 suites; added reproducible scenarios and evidence.
1.2 22 October 2026	Minh	Integrated prior Part A roles, client, risk priorities, deployment check, defect tracking and D3 milestone.
1.3 22 October 2026	Minh	Recorded client acceptance and feedback. Added new AI feature mapping and eight browser cases; retained existing outcomes.
1.4 22 October 2026	Minh	Replaced evidence paths with fixed GitHub links and published the missing source documents.
V1.5 — 24 October 2026	Minh	Aligned with eight customer use cases in final D4 pp. 62–72 and SRS; executed current tests, retained failures/retests and replaced provisional statuses with observed dispositions.
V1.6 — 24 October 2026	Minh	Numbered and titled all 26 tables, added linked references in the body text, and updated report authorship.

Report status: revised against all eight customer use cases, with current execution dispositions and disclosed coverage limits. Sarah Thomas is the testing lead and final technical reviewer. The release decision must account for the known intermittent deployed response, unidentified deployed source revision and administrator/browser-platform coverage gaps. No reviewer signature is asserted.

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The contents entries are internal links. Page numbers appear in the footer. Case identifiers remain stable when the document is merged into the final D4 PDF.

How to interpret results

The status definitions explain how to interpret each recorded test outcome (see TABLE 2).

TABLE 2 Test result status definitions

Status	Meaning
Pass	Observed output met the stated expected output for the identified test, build and environment.
Fail	The executed case produced a functional or assertion mismatch. Preserve the actual result and defect reference.
Blocked	A prerequisite prevented a valid execution or a dependent step could not be assessed.
Coverage gap	A required behaviour has no executed evidence here. It is disclosed explicitly and is not counted as a pass.
Historical Pass or Fail	A supplied source recorded that outcome. It is retained with its date and limitations; it does not establish a fresh result.
Feature gap	The desired capability is outside the implementation demonstrated by that run. Record separately from transport or setup failures.

One script may contain many assertions. One scenario may contain several turns and repeated runs. Script totals, case totals, turn checks, HTTP responses and human acceptance decisions use different denominators and are reported separately.

Part A Test plan

Scope and test objectives

The customer scope is Browse / Search Products (BR), Shop via Conversation (UC01), Clarify Shopping Preferences (UC02), Obtain Product Recommendations (UC03), Receive Complementary Product Recommendations (UC04), View Product Details (UC05), Refine Shopping Preferences (UC06) and Manage Shopping Cart (UC07). Accessory recommendations, cross-selling and upselling are UC04 branches. Team maintenance/testing and sponsor review/handover in Figure 7 are supporting activities, evaluated through regression, quality checks and the acceptance record.

The primary customer conversation route is POST /api/conversation-search. Original recommendation API cases target the older development-only POST /api/recommendations route, which intentionally returns HTTP 404 outside development. These layers are not interchangeable. [E4, E11]

Source of truth: final D4 Figure 7, page 62; use case descriptions, pages 63–72; SRS FR-01–FR-40, pages 42–46. The diagram shows eight customer use cases, but the text numbers only UC01–UC07. Table 50 also captions cart management as UC08 while its body says UC07. This report preserves the body identifiers and uses BR for the unnumbered Browse / Search Products case. The matrix below replaces the provisional TO-only mapping. [E17] (see TABLE 3).

TABLE 3 Customer use cases and test coverage

Use case	Behaviour under test	Requirement / source	Test coverage
BR — Browse / Search Products	Browse catalogue; search by product terms; recover from no results. Diagram-only customer use case.	Figure 7 p. 62; web experience	TC-BR-01–03; TC-PD-13; TC-REG search-filters
UC01 — Shop via Conversation	Enter natural-language requirements, obtain recommendations and continue the conversation; disclose failures.	FR-01–03, FR-25–29; pp.63–64	TC-PE-01–08; TC-AI-04; TC-HTTP-01–05; TC-INT-01/02
UC02 — Clarify Shopping Preferences	Ask only for necessary missing/unclear information; accept unknown optional preferences; use latest explicit answers.	FR-03–06, FR-18; pp.64–65	TC-PV-03–06; TC-AIX-01/06; S3-A/B; S6-A/B; guided-discovery suites
UC03 — Obtain Product Recommendations	Retrieve/filter/rank real products; explain matches and unmet soft preferences; preserve hard limits.	FR-07–14, FR-27, FR-30, FR-33; pp.66–67	TC-AI-01–10; TC-INT-03; TC-AIX-07; grounded/validation suites
UC04 — Receive Complementary Product Recommendations	Coordinate multiple roles; provide optional accessories/cross-sells/value alternatives; preserve applicable budget and consent.	FR-21–24; pp.67–68	TC-AIX-02–05/08; B1–B5; S4-A/B; S5-A–F; cross-sell/upsell/budget suites

Use case	Behaviour under test	Requirement / source	Test coverage
UC05 — View Product Details	Open selected product; display supported image, price, colour, size and availability; handle missing details/images.	FR-11, FR-13, FR-31, FR-33; pp.69–70	TC-PD-01–03/06/07/12; TC-UI-01; TC-AIX-07; TC-REG-47/48
UC06 — Refine Shopping Preferences	Change one stated preference, retain others, use newest explicit value, reject/replace recommendations or clarify ambiguity.	FR-15–20, FR-28; pp.70–71	S1-A–D; S2-A–D; S3-C; CAT/COL/SET; cheaper/scoped-state suites
UC07 — Manage Shopping Cart	Select size, add available item, inspect cart, change quantity/remove, retain cart and display empty state.	FR-32; pp.71–72 (Table 50 caption says UC08)	TC-PD-04/05/08–14; TC-UI-02; cart and failure boundary checks

Scope boundaries

A product can display its one recorded catalogue colour without supporting colour-variant switching. TC-PD-02 and TC-UI-01 now verify accurate fixed-colour presentation for the approved MVP; this is a specification revision, not evidence that the old variant interaction passed. Supplier composition, personal garment fit and per-size stock are not established by aggregate availability or listed sizes. Synthetic product specifications remain labelled. Real checkout, payment and order placement are excluded by the SRS. A complete security audit, sustained load/availability test and native-device browser matrix are not established by these checks.

Use case flow coverage and limits

The eight-case matrix establishes traceability. The following review maps the described alternative flows and identifies the limits of the execution evidence. A coverage gap is not counted as a pass (see TABLE 4).

TABLE 4 Alternative use case flows and coverage limits

Use case / scope	Flows addressed	Evidence disposition / limit
UC01 / UC02 / UC06	UC01 A1–A3: clarification, rephrasing and technical failure. UC02 A1–A4: unclear/conflicting/known/unknown answers. UC06 A1–A4: newest value, rejection, ambiguity and retained preferences.	PE/PV, guided-discovery, scoped-state and repeated S1–S3/S6 tests; AIX01/06; controlled INT01/02. Pass for recorded inputs, not arbitrary language.
UC03	A1–A2: soft alternatives and disclosed mismatches; A3: hard no-match; A4: browse without size/budget.	AI06–10, AIX06 and DEP no-match checks Pass. No catalogue-wide human relevance audit is claimed.
UC04	A1: coordinated roles/budget; A2: cross-sell; A3–A4: value/upsell with budget consent; A5: no eligible extra.	AIX02–05/08 and budget/cross-sell suites. AIX04 has intermittent DF6; AIX08 fails saved-budget/value comparison (DF8). No-eligible-extra domain assertions Pass.
UC05	A1: changed availability; A2: incomplete data; A3: retrieval error; A4: missing-image placeholder.	Details/navigation and fixed-colour browser checks Pass; incomplete facts and image fallbacks have offline coverage. Live availability mutation and product-page retrieval-error UI are coverage gaps.
UC07	A1: remove; A2: availability changes before add; A3: technical cart failure; A4: empty cart.	Remove/empty state, size selection, quantity and persistence Pass on desktop/mobile. Availability race and client storage failure alternatives lack injected execution evidence.

Testing strategy and resources

Strategy applied to the product

Tests follow the revised use case main flows, alternatives and failure conditions. Priorities are explicit size/budget/exclusion constraints, catalogue-supported explanations, correct multi-turn updates and purchase totals, then UI presentation. Each result identifies its execution layer: deterministic fixture, real model with synthetic catalogue, live read-only catalogue, actual HTTP route, deployed browser, or client decision. An assertion at one layer does not establish all the others (see TABLE 5).

TABLE 5 Testing strategy by execution layer

Layer and approach	Execution and purpose	Resource and output
Unit and contract White box	Run deterministic preference, schema, price arithmetic, grounding and state checks against defined fixtures. Inspect assertions and branches that enforce constraints, unknown values and invalid transitions.	Developer responsible for module; Sarah reviews outcomes. Saved assertion output and source hashes.
Domain integration White and black box	Exercise makeTurn, interpretation plan handling, catalogue filtering, finishConversationTurn and acceptReply together. Controlled plans isolate server behaviour from model variability.	Existing offline scripts and synthetic fixtures. JSON/session checks and regression log.
HTTP integration Black box	Exercise the real conversation route and the legacy development route separately. Check content type, JSON limits, status/body contracts, external failures and response confidentiality.	Developer plus Sarah on isolated test configuration. Sanitised request/response records.
Browser and responsive system Black box	Use a fresh browser session, named product fixture and declared viewport. Exercise product selection, bag count, quantities, summaries, navigation, saved state and visible errors.	Sarah or delegate; desktop Chrome and a mobile browser, with versions recorded. Screenshots and execution sheet.
Real model customer journeys Black box	Start each scenario independently, use saved catalogue fixtures and run the intended messages in order. Record every repetition and dependent step, including blocked or variable outcomes.	Configured model and test catalogue. Transcripts, interpreted plans, product IDs and expected-versus-actual checks.
Regression and confirmation	After each repair rerun the failing case and affected suites; rerun all 48 offline scripts before freezing the candidate. Recheck high-risk customer journeys on that same candidate.	Developer produces fix evidence; Sarah confirms closure. Old failures retained beside the retest.
Acceptance and handover	Retain the 21 October acceptance decision, feedback and team response. Confirm the final candidate and verify post-meeting changes before handover.	Peyton Bernhardt and all team members attended. Salvi supplied the acceptance record; Sarah completes the build link. [E13]

Tools and environment

The final offline run used Node v24.14.0 on macOS; earlier scenario summaries record Node v24.22.1. The checkout uses Next.js 16.3.8, React 19.2.8 and the installed lockfile dependencies. The candidate started at Git HEAD 85d4f1b5e0a22e5dd86445005ebb585055eae7e0 on feat/stylist-ui and includes the retained matching-set/budget, search recovery and harness changes. E14 supplies the source manifest, patch, snapshots and quality-check output; HEAD alone is not the tested candidate.

Live extraction used the configured OpenAI model; catalogue checks used a read-only public Supabase client and a saved snapshot. Browser actions used the embedded browser at recorded viewports of 1440 × 900, 1280 × 720 and 390 × 844. Its engine version was not exposed by the tool; no Chrome/Safari/Edge or physical iOS/Android certification is claimed. Evidence captures timestamps, exact screenshot dimensions, product IDs and visible state without credentials (see TABLE 6).

TABLE 6 Testing tools and environments

Tool or environment	Use and evidence boundary
Supabase dashboard	Read-only public Supabase client retrieved the live snapshot. No shared catalogue record was changed; no dashboard CRUD execution is claimed.
Desktop and mobile developer tools	Browser screenshots, accessibility state and captured console logs accompany observed actions. The embedded browser engine version was not exposed.
Vercel deployment	Confirmed submission URL https://abcandco.vercel.app/: HTTP and browser behaviour observed. Source commit/deployment equality with local fixes is unverified.
GitHub pull requests and Jira	Retain the prior plan tool choices [E12] for defect owners, code review and fix verification. Link an existing issue and pull request where used; keep the report defect ID as a stable reference.

Team responsibilities

Minh prepared this report and compiled the scripted and browser test evidence from 24 October. Sarah Thomas is the confirmed testing lead and independent technical reviewer. Report preparation and evidence compilation are separate from final team review and approval (see TABLE 7).

TABLE 7 Team roles and responsibilities

Role	Team member	Responsibility
Testing lead and QA	Sarah Thomas	Own planning, allocations, final SRS coverage, documentation, evidence review, defect disposition and client coordination. Assign a second reviewer for critical budget, constraint and bag checks.
Frontend developers	Salvi, Hans, Taylor	Prepare fixtures and run functional, UI and compatibility checks for their components. Capture desktop/mobile behaviour, network and console errors, bag arithmetic and persistence evidence.
AI and recommendation leads	Beatrice, Abrar	Verify extraction, recommendation quality, hard constraints, multi-turn state and AI integration. Retain model scenario inputs, transcripts, product IDs and fix evidence.
Data and testing leads	Sarah Thomas, Taylor, Abrar	Prepare test data and saved catalogue snapshots; support planning, HTTP integration, controlled failures and overall system testing. Sarah confirms individual delegates.
Client representative	Peyton Bernhardt Salesforce	Attended the 21 October meeting with all team members. Product accepted as ready; further proactive recommendations suggested while the current version was satisfactory. [E13]
Automated regression runner	Node scripts with the actual operator recorded separately	Automated scripts retain timestamps and assertion output. Minh compiled the execution evidence; Sarah remains the testing lead and independent technical reviewer.

Execution accountability

For manual cases, record the person who performed the actions. For automated cases, record the script or harness, its version or source hash, and the person or agent who launched and supervised the run as separate fields. A reviewer records confirmation and closure separately from the developer who supplied the fix.

Test schedule and responsibilities

D4 is due on 24 October 2026. Dates use Australia/Sydney; raw tool evidence retains UTC timestamps (AEDT = UTC + 11 hours). The schedule below is an execution record with review ownership. Confirm the institutional upload time independently.

The prior plan records the D3 Test Plan and initial Test Case Specifications milestone on 1 October 2026 [E12]. Remaining D4 work follows below (see TABLE 8).

TABLE 8 Testing schedule and review responsibilities

Date	Work and milestone	Owner and resources	Deliverable and state
21 Oct 2026	Client demonstration, approval and feedback.	Peyton Bernhardt and all team members; Salvi supplied acceptance/minutes.	Accepted; final D4 records meeting at 6:00 pm. Demonstrated source revision not supplied.
22 Oct 2026	Earlier report and 46-script baseline.	Minh — baseline evidence compilation; Sarah is accountable for team review.	Historical evidence retained as E0–E13; superseded by current execution results.
24 Oct 2026	Run all 38 real-model customer scenarios three times; retain failures and retests.	Minh — model test evidence compilation; configured model and synthetic catalogue.	Full run 108 Pass / 6 provider timeouts; affected scenarios repeated three times, 9/9 Pass. All attempts retained.
24 Oct 2026	Execute product/cart, search and AI controls on deployed desktop/mobile browser.	Minh — browser evidence compilation; saved screenshots, accessibility state and logs.	Observed results recorded; DF6–DF8 retained. Search recovery fixed and retested locally.
24 Oct 2026	Execute catalogue/model/HTTP and controlled dependency failures; run final quality checks.	Minh — integration and quality-check evidence compilation; read-only Supabase and isolated HTTP processes.	32 latest checks Pass; 3 failure/recovery checks Pass; 48 suites, typecheck, lint and build Pass.
24 Oct 2026	Align report to eight use cases and final SRS; export and inspect evidence links.	Minh — report preparation; Sarah remains the final team reviewer.	Revised report and evidence package produced. Known failures and unproved coverage remain explicit.
24 Oct 2026	Final technical decision and institutional submission.	Sarah Thomas / designated team submitter.	An all-passing technical decision is not supported while DF6–DF8 and the stated coverage limits remain unresolved. No submission or review signature is asserted.

Testing milestones

The 21 October client decision remains Accepted. The 24 October technical run, fixes, retests and document/use-case reconciliation are recorded in E14–E17. Earlier milestones remain historical; they are not backdated execution claims.

Test control and completion criteria

Entry criteria and reproducible setup

- Record Git HEAD and all modified source hashes. E14 contains the initial and final local manifests and source patch. Local execution and deployed URL observations have distinct candidate identities; the Vercel source commit was not exposed.
- Install from the lockfile in a clean environment when reproducing. Record Node and dependency versions. Do not repair or overwrite an existing active checkout merely to prepare documentation.
- For offline cases, use the checked-in synthetic fixture definitions. For live cases, save a read-only catalogue snapshot and exact product IDs, price, size labels and availability prerequisites.
- For UI cases, start with an empty bag and a New chat unless the case states otherwise. Record browser, version, viewport and persisted state. Use a separate browser profile for isolation.
- For any service-failure test, use an isolated process or controlled test adapter. Do not invalidate shared credentials or interrupt the shared database.

Execution commands and conditions

The execution commands identify the method and conditions for reproducing each test layer (see TABLE 9).

TABLE 9 Execution commands and conditions

Purpose	Command or method	Limits
Offline full run	npm test	Uses scripts/run-regression.mjs. No live/API/browser coverage. Stop and investigate any missing setup dependency.
Single offline suite	node --import tsx scripts/FILE	Substitute the exact suite filename from Part B. Add --conditions=react-server for grounded-recommendation, conversation-production and stylist-telemetry. The production suite also needs NODE_ENV=production.
Real-model in-process scenarios	npm run test:scenarios	Uses configured model credentials and synthetic catalogue; three repetitions per scenario were executed. Save every outcome. It does not test HTTP or rendering.
Legacy API validation	node scripts/test-recommendation-api-validation.mjs	Requires development server at http://127.0.0.1:3000. Development-only recommendation route; does not certify the customer conversation route.
Build and static checks	npm run typecheck; npm run lint; npm run build -- --webpack	Separate checks, all completed with exit code 0 on 24 October after the local search recovery fix. See E14 logs.
Browser and client tests	Execute specified actions on the frozen candidate; save results and evidence	21 October client decision is in E13. Browser cases require separate observed results; screenshots support their captured interaction.

Completion and acceptance criteria

Recommend technical completion only when prioritised MVP requirements have mapped tests and observed dispositions on the intended candidate; all 48 offline suites pass after relevant changes; and no unresolved critical/high constraint, grounding, arithmetic or failure-disclosure defect remains. Use case mapping is complete for all eight customer cases. Full alternative-flow and SRS compliance require the coverage limitations below to be explicitly reviewed, not silently converted to passes.

These criteria guide Sarah's technical completion review. Peyton's product acceptance on 21 October is recorded in E13. The meeting record does not supply individual case outcomes or establish that every technical completion criterion has been met.

Defect handling and execution record

Each record retains the case ID, execution method, date and time, input, expected and actual result, layer, candidate, evidence and defect reference. Minh compiles the report and evidence; independent team review and defect closure are separate. DF identifiers are report references, not Jira keys.

Treat constraint violations, fabricated products, incorrect purchase totals or disclosure of secrets as high priority; an unavailable required journey is at least high until triaged. Record cosmetic issues separately. Preserve original failures. Move each defect through recorded triage, owner allocation, fix review, dated confirmation and affected regression checks before Sarah closes it or records an accepted exception.

Risks and contingencies

The risk register records testing constraints and the response assigned to each risk (see TABLE 10).

TABLE 10 Testing risks and contingencies

Risk	Response and owner
Changing source or live catalogue	Freeze a candidate and retain hashes/snapshot. Sarah distinguishes source, catalogue and model variation.
Missing dependencies or service credentials	Mark Blocked with exact setup failure. Developer restores an isolated environment; Sarah reschedules without claiming a pass.
Time available before 24 October	Prioritise core constraints, combined budgets, bag flows and honest errors. Sarah records deferred cases and their impact.
Unidentified demonstration build	Keep the 21 October acceptance decision; Sarah records the demonstrated URL/build and verifies the final candidate and post-meeting changes.
Single colour and per-size stock limits	Test the actual recorded colour and listed sizes. The revised fixed-colour case does not establish variant switching or stock for a particular size.

Part B Original functional cases

All 45 original test identifiers are retained. The result column now records 24 October observations; historical outcomes remain in the original source E0. PD colour-variant switching is explicitly replaced by fixed-colour display because the revised UC05 requires catalogue-supported details and the implemented product has one colour. No claim of variant-switch support is made.

Current browser fixture PD1: live catalogue product P1031, Gold Trench Coat, Gold, A$90, sizes XS/S/M/L/XL at https://abcandco.vercel.app/products/P1031. Start with an empty bag. Different sizes of the same product are separate rows; repeated same-size adds increase that row quantity. The recorded desktop/mobile runs use this fixture rather than the earlier mock blouse.

Product details and bag

The product and cart cases specify the inputs, expected behaviour and observed outcomes (see TABLE 11).

TABLE 11 Product details and shopping cart test cases

Case and purpose	Exact input or action for next execution	Expected result	Observed result and status
TC-PD-01 Product details load (UC05)	Open /products/P1031 (PD1) on desktop; compare details with the saved fixture record.	Page displays product image(s), name, price, colours and sizes from the catalogue	Pass — P1031 details loaded; Gold, A$90 and XS/S/M/L/XL matched the live fixture. [24 Oct evidence]
TC-PD-02 — Recorded colour display (UC05; scope revised)	Open P1031; compare the Gold catalogue colour with the visible swatch/label and saved cart row. Do not require nonexistent variants.	The one recorded colour is displayed accurately and consistently; the UI makes no unsupported variant promise.	Pass — revised fixed-colour test: Gold label/swatch matched P1031. One catalogue colour exists; colour switching is outside this case's revised MVP scope. [24 Oct evidence]
TC-PD-03 Size selection (UC05)	On PD1 click size S, then M. Record selected state after each click.	Selected size is highlighted	Pass — selected S and then M; selection state changed correctly. [24 Oct evidence]
TC-PD-04 Add selected variant to bag (UC07)	On P1031 select S and Add to bag; inspect Gold/S and A$90 in the cart.	Item is added to the cart with the selected colour and size	Pass — selected S, added P1031; bag showed Gold/S at A$90. [24 Oct evidence]
TC-PD-05 Reject missing size (UC07)	On fresh PD1 do not select a size; activate Add to bag and inspect prompt and bag count.	Item is not added; user is prompted to select a size	Pass — Add without size displayed "Please select a size"; bag count remained zero. [24 Oct evidence]

Case and purpose	Exact input or action for next execution	Expected result	Observed result and status
TC-PD-06 Expand and collapse description (UC05)	On PD1 open the description section, then activate it again.	Section expands to show details; clicking again collapses it	Pass — Details expanded to recorded construction/fit/style/occasion information and collapsed on the next activation. [24 Oct evidence]
TC-PD-07 Mobile product details (UC05)	Open PD1 at a 390 by 844 mobile viewport; record device/browser version. Repeat selection and inspect overflow and controls.	Layout adjusts to the screen without overlapping or cut-off content	Pass — 390 × 844 product view retained readable details, size controls and Add action without blocking overlap. [24 Oct evidence]
TC-PD-08 Keep different sizes as separate bag rows (UC07)	Add one PD1 Gold S and one PD1 Gold M. Expected subtotal A$180 before any disclosed adjustments.	The cart shows two separate rows for the same product, one for size S and one for size M. The Order Summary updates to reflect both items.	Pass — S quantity 1 and M quantity 2 were separate rows; total units 3 and subtotal A$270. [24 Oct evidence]
TC-PD-09 Repeated add increases quantity (UC07)	Add PD1 Gold M twice. Expected quantity 2, total units 2 and item subtotal A$180.	Its quantity increases to 2. The Shopping Cart number and Order Summary updates.	Pass — repeated M add increased its row quantity to2; S1+M2 total A$270. [24 Oct evidence]
TC-PD-10 Remove bag item (UC07)	With one PD1 Gold M in the bag, remove it. Expected count 0 and item subtotal A$0.	The item disappears. The Shopping Cart number and Order Summary recalculate.	Pass — removing S left M2, bag count 2 and A$180; final mobile removal also produced the empty-cart state. [24 Oct evidence]
TC-PD-11 Adjust bag quantity (UC07)	With PD1 quantity 1, press plus then minus. Expected quantities 2 then 1 and item subtotals A$180 then A$90.	The quantity changes by one with each click. The Shopping Cart number and Order Summary updates accordingly.	Pass — M quantity 2→3→2 recalculated A$180→270→180. Reload retained product, M and quantity 2; mobile persistence also verified. [24 Oct evidence]
TC-PD-12 Open product from bag (UC07)	From the bag click PD1; expect /products/P1031.	The corresponding product details page opens.	Pass — P1031 link from bag opened the matching Gold Trench Coat product page at A$90. [24 Oct evidence]
TC-PD-13 Continue browsing (UC07)	From the bag activate Continue Browsing Collection; record the destination and compare with the current link target.	The product browsing page opens.	Pass — Continue Browsing opened the collection/homepage and retained the bag. [24 Oct evidence]
TC-PD-14 Open styling advice link (UC07)	From the bag activate the styling advice or concierge link; compare the opened destination with the current link target. Record changed/removed links as a requirement review item.	The contextual AI Stylist interface opens for further shopping assistance.	Pass — cart styling-advice control opened AI Stylist in context. [24 Oct evidence]

Recommendation and API behaviour

TC-AI-01–10 use the 24 October live read-only catalogue snapshot: 318 records, including 316 available products. Counts are observations, not fixed acceptance targets. TC-AI-11–14 validate the development-only legacy endpoint; the deployed legacy endpoint correctly returns 404. The production conversation route has separate HTTP cases. [E14] (see TABLE 12).

TABLE 12 Recommendation and API test cases

Case and purpose	Exact input or action for next execution	Expected result	Observed result and status
TC-AI-01 Product catalogue retrieval	Request catalogue products from Supabase	System retrieves available catalogue products successfully	Pass — read 318 current catalogue records (316 available); snapshot retained. [24 Oct evidence]
TC-AI-02 Hard filtering by size and budget	Size = 10, max price = $100	Only available, in-stock products containing size 10 and costing <= $100 are returned	Pass — 60 available/in-stock records met size 10 and maximum A$100. [24 Oct evidence]
TC-AI-03 Soft preference ranking	Size 10, max $100, category Clothing, preferred colour Lavender	Products passing hard constraints are ranked by preference match	Pass — fresh live catalogue/model/domain execution met the stated assertion. [24 Oct evidence]
TC-AI-04 Full recommendation pipeline	“Show me lavender clothing for a beach wedding, with a relaxed style, in size 10, costing no more than $100 per item.”	Preferences are extracted, products filtered/ranked, and best recommendation returned	Pass — fresh live catalogue/model/domain execution met the stated assertion. [24 Oct evidence]
TC-AI-05 Maximum budget boundary	Lavender clothing, size 10, maximum price $75	Products above $75 are excluded; products at exactly $75 remain valid	Pass — fresh live catalogue/model/domain execution met the stated assertion. [24 Oct evidence]
TC-AI-06 Fallback alternatives when no exact match	Lavender clothing, size 10, max price $50	Hard constraints remain enforced; soft preferences may be relaxed and unmatched preferences identified	Pass — fresh live catalogue/model/domain execution met the stated assertion. [24 Oct evidence]
TC-AI-07 Explicit colour exclusion	Lavender clothing, size 10, max $50, exclude Brown	Brown products must not be returned, even as alternatives	Pass — fresh live catalogue/model/domain execution met the stated assertion. [24 Oct evidence]
TC-AI-08 Required size enforcement	Lavender clothing, size 12, max $50	Products not available in size 12 must be excluded	Pass — fresh live catalogue/model/domain execution met the stated assertion. [24 Oct evidence]

Case and purpose	Exact input or action for next execution	Expected result	Observed result and status
TC-AI-09 Occasion and style ranking	Clothing, Lavender, Beach Wedding, Relaxed, size 10, max $100	Product matching all four soft preferences should rank as an exact match	Pass — fresh live catalogue/model/domain execution met the stated assertion. [24 Oct evidence]
TC-AI-10 No product satisfies hard constraints	Size 10, max price $1	No unsuitable or over budget product should be returned; response should indicate no match	Pass — no eligible products at maximum A$1; hard-filter result was empty. [24 Oct evidence]
TC-AI-11 Empty/invalid shopping request handling	POST /api/recommendations separately with application/json bodies {"message":" ","limit":6} and {"limit":6}.	API rejects invalid input without processing recommendations. Error body identifies INVALID_REQUEST, success false and a nonempty request_id. See scripts/test-recommendation-api-validation.mjs.	Pass — current development HTTP validation run returned the expected 400/415 and controlled error contract. Production legacy endpoint was separately confirmed 404. [24 Oct evidence]
TC-AI-12 Invalid recommendation limit	POST /api/recommendations with application/json and {"message":"Show me clothing.","limit":7}. Valid upper boundary limit 6 is checked by TC-REG-15.	API rejects values above the supported limit of 6. Error body identifies INVALID_REQUEST, success false and a nonempty request_id. See scripts/test-recommendation-api-validation.mjs.	Pass — current development HTTP validation run returned the expected 400/415 and controlled error contract. Production legacy endpoint was separately confirmed 404. [24 Oct evidence]
TC-AI-13 Malformed JSON request	POST /api/recommendations with Content-Type application/json and exact malformed body {"message":	API returns controlled validation error instead of crashing. Error body identifies INVALID_REQUEST, success false and a nonempty request_id. See scripts/test-recommendation-api-validation.mjs.	Pass — current development HTTP validation run returned the expected 400/415 and controlled error contract. Production legacy endpoint was separately confirmed 404. [24 Oct evidence]
TC-AI-14 Incorrect content type	POST /api/recommendations with Content-Type text/plain and body {"message":"Show me clothing."}.	API rejects unsupported content type. Error body identifies INVALID_REQUEST, success false and a nonempty request_id. See scripts/test-recommendation-api-validation.mjs.	Pass — current development HTTP validation run returned the expected 400/415 and controlled error contract. Production legacy endpoint was separately confirmed 404. [24 Oct evidence]
TC-AI-15 AI service unavailable	Simulate OpenAI/API failure	System should return an understandable error without exposing sensitive details	Pass — actual POST route in an isolated HTTP process returned 502 CONVERSATION_FAILED; safe message, no fabricated products or shared-service disruption. See TC-INT-01/02. [24 Oct evidence]
TC-AI-16 Database unavailable	Simulate Supabase/databas e failure	System should return an appropriate catalogue unavailable error without returning fabricated product results.	Pass — actual POST route in an isolated HTTP process returned 503 CATALOGUE_UNAVAILABLE; safe message, no fabricated products or shared-service disruption. See TC-INT-01/02. [24 Oct evidence]

Preference extraction

The extraction cases check stated preferences and preserve values that the customer has not supplied (see TABLE 13).

TABLE 13 Preference extraction test cases

Case and purpose	Exact input or action for next execution	Expected result	Observed result and status
TC-PE-01 Extract dress, occasion, colour, size and budget	“I need a black dress under $150 for a wedding in size 10.”	Structured preferences identify Clothing, Dresses, Wedding Guest, Black, size 10 and maximum price $150	Pass — 24 October fresh real-model extraction matched all specified fields and preserved unspecified values. [24 Oct evidence]
TC-PE-02 Avoid inferring an unstated product category	“I’m going to a beach wedding and want something elegant, maybe sage or blue.”	Extract Beach Wedding, Elegant, Sage and Blue without assuming a dress, category, gender or season	Pass — 24 October fresh real-model extraction matched all specified fields and preserved unspecified values. [24 Oct evidence]
TC-PE-03 Extract a casual brunch request	“I need something casual for brunch under $100.”	Extract Brunch, Casual and maximum price $100 without assuming a product category	Pass — 24 October fresh real-model extraction matched all specified fields and preserved unspecified values. [24 Oct evidence]
TC-PE-04 Extract gender, category and size	“Show me men’s shoes for work in size 11.”	Extract Men, Shoes, Work and size 11 without inventing a shoe subcategory	Pass — 24 October fresh real-model extraction matched all specified fields and preserved unspecified values. [24 Oct evidence]
TC-PE-05 Extract an excluded colour	“I want a romantic outfit for date night but nothing red.”	Extract Romantic and Date Night; put Red in excluded colours, not preferred colours	Pass — 24 October fresh real-model extraction matched all specified fields and preserved unspecified values. [24 Oct evidence]
TC-PE-06 Extract minimum and maximum prices	“A bag costing at least $100 and up to $150.”	Extract Bags, minimum price $100 and maximum price $150	Pass — 24 October fresh real-model extraction matched all specified fields and preserved unspecified values. [24 Oct evidence]
TC-PE-07 Reject empty extraction input	Call extractPreferences with the exact whitespace-only string " ". Confirm rejection before any OpenAI call.	Function rejects the request before calling OpenAI	Pass — 24 October fresh blank-input/HTTP validation returned the specified controlled rejection. [24 Oct evidence]
TC-PE-08 Reject invalid extraction API bodies	POST /api/extraction independently with application/json bodies A: {"message": (malformed), B: {}, C: {"message":7}, D: {"message":" "}. Record each response separately.	API returns HTTP 400 with a controlled validation error	Pass — 24 October fresh blank-input/HTTP validation returned the specified controlled rejection. [24 Oct evidence]

Preference validation

TC-PV-01–06 passed within the fresh deterministic validation suite on 24 October. TC-PV-07 separately queried the live Supabase vocabulary. Deterministic and live outcomes are labelled separately. [E14] (see TABLE 14).

TABLE 14 Preference validation test cases

Case and purpose	Exact input or action for next execution	Expected result	Observed result and status
TC-PV-01 Normalise recognised catalogue values	Style “ elegant ” and preferred colours [" navy ", "NAVY"]	Return catalogue spelling Elegant and one Navy, without changing the original input	Pass — fresh deterministic validation assertions passed in the 48-suite run. [24 Oct evidence]
TC-PV-02 Flag an unsupported soft preference	Occasion Brunch and style Casual	Preserve both values and report Casual as an unsupported catalogue style without blocking the request	Pass — fresh deterministic validation assertions passed in the 48-suite run. [24 Oct evidence]
TC-PV-03 Flag an unknown size	Size “99”	Preserve the requested size, report it as unsupported and require clarification	Pass — fresh deterministic validation assertions passed in the 48-suite run. [24 Oct evidence]
TC-PV-04 Flag an unknown excluded colour	Exclude “Ultraviolet”	Preserve the exclusion and require clarification so retrieval cannot silently ignore it	Pass — fresh deterministic validation assertions passed in the 48-suite run. [24 Oct evidence]
TC-PV-05 Detect a category mismatch	Category Shoes with subcategory Dresses	Report that Dresses does not belong to Shoes and require clarification	Pass — fresh deterministic validation assertions passed in the 48-suite run. [24 Oct evidence]
TC-PV-06 Detect a conflicting budget	Minimum price $200 and maximum price $100	Report the invalid price range and require clarification	Pass — fresh deterministic validation assertions passed in the 48-suite run. [24 Oct evidence]
TC-PV-07 Validate against the live Supabase vocabulary	Occasion Brunch and style Casual, using the public catalogue connection	Recognise Brunch and flag Casual as an unsupported style	Pass — fresh live vocabulary recognised Brunch and retained a non-blocking unsupported Casual style issue. [24 Oct evidence]

Browse / Search Products — eighth customer use case

BR is a report alias for the unnumbered customer function in Figure 7. These cases are separate from conversational discovery and retain their own browser evidence (see TABLE 15).

TABLE 15 Browse and search product test cases

Case / use case	Input / action	Expected output	Actual output
TC-BR-01 — product search	Open Shop; query “Gold Trench Coat”.	Matching catalogue cards and links; accurate names/prices; bag unchanged.	Pass — P1031 A$90 and P1093 A$215 returned on mobile. [E16]
TC-BR-02 — no results	Search “zzzd4nomatch2026”.	Zero products, understandable no-results message and recovery action.	Pass — 0 products; “No products found”; broader-term guidance and Browse all products link.[E16]
TC-BR-03 — recovery	Activate Browse all products after no results.	Return to the unfiltered collection with selectable product cards.	Fail deployed: query/results remained. Pass local retest after fix: cleared query, 316 products and restored cards. DF7; deployment not established.[E16]

Current regression suite specifications

Fresh final execution: 24 October 2026, 18:34:03–18:34:11 AEDT (07:34:03–07:34:11 UTC). All 48 scripts passed after the search recovery change. The source was unchanged during the run. E14 records source hashes and local changes relative to HEAD 85d4f1b. Typecheck, lint and the production webpack build also completed successfully.

TC-REG-01–46 retain their identities and historical source links. TC-REG-47 and 48 identify the new image-delivery and product-image suites. E14 contains the exact current script/fixture snapshots used on 24 October; use those for current assertions rather than assuming the older E10 source snapshot describes later edits (see TABLE 16).

TABLE 16 Automated regression suite specifications

Identifier	Script on GitHub	Input fixture and expected assertion reference	Fresh result
TC-REG-01	test-stylist-remediation.mjs	Example assertion group lines 29 to 33: A trailing show-options phrase retains a complete mens shirt brief and canonical medium size. Full input and assertion definitions: lines 1 to 148; 18 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-02	test-conversation-recovery.mjs	Example assertion group lines 34 to 40: No size filter clears an invalid size without inventing filter. Full input and assertion definitions: lines 1 to 140; 14 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-03	test-matching-set-budget.mjs	Example assertion group lines 37 to 45: The exact prompt immediately searches two groups under one A$500 total. Full input and assertion definitions: lines 1 to 136; 14 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-04	test-category-collection-defaults.mjs	Example assertion group lines 30 to 38: The exact dress prompt searches Women immediately with Work and A$200. Full input and assertion definitions: lines 1 to 193; 18 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-05	test-collection-first.mjs	Example assertion group lines 31 to 38: The exact request asks collection first and saves the full clothing brief. Full input and assertion definitions: lines 1 to 184; 18 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]

Identifier	Script on GitHub	Input fixture and expected assertion reference	Fresh result
TC-REG-06	test-product-card-presentation.mjs	Example assertion group lines 13 to 16: A lone clothing size still requires an explicit choice. Full input and assertion definitions: lines 1 to 61; 10 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-07	test-guided-discovery.mjs	Example assertion group lines 27 to 37: The screenshot's uncertain request asks one easy question without inventing preferences. Full input and assertion definitions: lines 1 to 223; 20 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-08	test-search-filters.mjs	Example assertion group lines 13 to 18: Audited catalogue and narrowing journey match real prototype facts. Full input and assertion definitions: lines 1 to 95; 17 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-09	test-selected-outfit-review.mjs	Example assertion group lines 38 to 44: Common current-outfit opinions, including the supplied wording, use the review intent. Full input and assertion definitions: lines 1 to 169; 18 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-10	test-outfit-interface.mjs	Example assertion group lines 37 to 41: Selected prices total exactly and leave A$15; owned items cost zero. Full input and assertion definitions: lines 1 to 141; 15 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-11	test-grouped-discovery-memory.mjs	Example assertion group lines 27 to 38: Plentiful inventory shows exactly five per group and every combination fits the shared budget. Full input and assertion definitions: lines 1 to 196; 17 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-12	test-outfit-budget.mjs	Example assertion group lines 31 to 40: Exact customer request returns a complete formal set at the inclusive A$400 boundary. Full input and assertion definitions: lines 1 to 142; 17 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-13	test-customer-scenario-regressions.mjs	Example assertion group lines 32 to 40: An explicit occasion survives even if the model omits it from an otherwise clear broad brief. Full input and assertion definitions: lines 1 to 227; 21 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]

Identifier	Script on GitHub	Input fixture and expected assertion reference	Fresh result
TC-REG-14	test-preference-validation.ts	Exact case array, fixture values and assertions: lines 1 to 39. Assertion statements start at line 14. Complete input/output checks are included in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-15	test-recommendation-contract.ts	Exact case array, fixture values and assertions: lines 1 to 80. Assertion statements start at line 55. Complete input/output checks are included in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-16	test-grounded-recommendation.ts	Exact case array, fixture values and assertions: lines 1 to 38. Assertion statements start at line 19. Complete input/output checks are included in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-17	test-validation-search.mjs	Example assertion group lines 80 to 101: Recognised labels are normalised without changing the input. Full input and assertion definitions: lines 1 to 224; 14 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-18	test-guided-follow-ups.mjs	Example assertion group lines 66 to 71: Cheaper asks for a price without inventing a budget. Full input and assertion definitions: lines 1 to 192; 14 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-19	test-cross-sell.mjs	Example assertion group lines 91 to 102: Recorded category returns a real complementary item. Full input and assertion definitions: lines 1 to 298; 15 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-20	test-cross-sell-client.mjs	Example assertion group lines 69 to 75: Additional budget and size are separate from the main request. Full input and assertion definitions: lines 1 to 222; 10 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-21	test-upsell.mjs	Example assertion group lines 112 to 128: Higher-priced replacement needs a specific added match. Full input and assertion definitions: lines 1 to 321; 16 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]

Identifier	Script on GitHub	Input fixture and expected assertion reference	Fresh result
TC-REG-22	test-upsell-client.mjs	Example assertion group lines 54 to 65: Request preserves all preferences without mutating them. Full input and assertion definitions: lines 1 to 237; 12 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-23	test-chat-request.mjs	Example assertion group lines 34 to 53: A successful request keeps the submitted preferences. Full input and assertion definitions: lines 1 to 190; 9 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-24	test-optional-discovery.mjs	Example assertion group lines 99 to 110: Cross-sell works without asking for size or budget. Full input and assertion definitions: lines 1 to 250; 8 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-25	test-card-actions.mjs	Example assertion group lines 77 to 84: The card supplies its product ID; missing budget and size stay null. Full input and assertion definitions: lines 1 to 188; 10 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-26	test-cheaper-search.mjs	Example assertion group lines 93 to 104: Recognises common standalone cheaper requests. Full input and assertion definitions: lines 1 to 293; 12 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-27	test-viewed-product.mjs	Example assertion group lines 47 to 56: The page exposes only the defined public product fields. Full input and assertion definitions: lines 1 to 244; 12 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-28	test-conversation-foundation.mjs	Example assertion group lines 55 to 63: The second product uses the displayed order. Full input and assertion definitions: lines 1 to 339; 18 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-29	test-conversation-plan.mjs	Example assertion group lines 95 to 107: A new search sets only supplied item preferences. Full input and assertion definitions: lines 1 to 431; 16 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]

Identifier	Script on GitHub	Input fixture and expected assertion reference	Fresh result
TC-REG-30	test-conversation-search.mjs	Example assertion group lines 173 to 184: A new search returns catalogue products without size or budget. Full input and assertion definitions: lines 1 to 492; 16 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-31	test-plan-evidence.mjs	Example assertion group lines 68 to 82: Valid evidence needs only one generation. Full input and assertion definitions: lines 1 to 177; 6 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-32	test-plan-recovery.mjs	Example assertion group lines 28 to 47: The model schema binds list, number and enum values to their fields. Full input and assertion definitions: lines 1 to 233; 10 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-33	test-conversation-answers.mjs	Example assertion group lines 37 to 48: Recommendation answers identify the requested product and factual occasion match. Full input and assertion definitions: lines 1 to 328; 27 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-34	test-conversation-production.mjs	Example assertion group lines 33 to 37: The customer conversation route accepts a valid request in production. Full input and assertion definitions: lines 1 to 93; 12 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-35	test-stylist-commerce.mjs	Example assertion group lines 33 to 38: Changing the anchor changes rankings on an occasion/style counterfactual. Full input and assertion definitions: lines 1 to 255; 31 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-36	test-stylist-feature-import.mjs	Example assertion group lines 11 to 15: The real starter pack remains unknown and produces no import. Full input and assertion definitions: lines 1 to 57; 8 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-37	test-product-specifications.mjs	Example assertion group lines 12 to 15: Every saved generated specification satisfies category and identity rules. Full input and assertion definitions: lines 1 to 43; 5 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]

Identifier	Script on GitHub	Input fixture and expected assertion reference	Fresh result
TC-REG-38	test-stylist-telemetry.mjs	Example assertion group lines 17 to 17: Telemetry is opt-in and performs no database writes when disabled. Full input and assertion definitions: lines 1 to 46; 10 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-39	test-conversation-integration.mjs	Example assertion group lines 44 to 50: New searches do not require size or budget. Full input and assertion definitions: lines 1 to 253; 25 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-40	test-shopping-presentation.mjs	Example assertion group lines 48 to 57: A single exact product uses its real name and price in the message. Full input and assertion definitions: lines 1 to 243; 32 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-41	test-friendly-shopping-message.mjs	Example assertion group lines 22 to 28: Multiple products get paragraphs with the real displayed price range. Full input and assertion definitions: lines 1 to 84; 12 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-42	test-reactive-demo-fixes.mjs	Example assertion group lines 31 to 35: Compound collection answer survives a later shopping clause. Full input and assertion definitions: lines 1 to 167; 22 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-43	test-stylist-journey-harness.mjs	Example assertion group lines 14 to 16: Independent family checks reject trousers offered as skirts. Full input and assertion definitions: lines 1 to 217; 20 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-44	test-journey-remediation.mjs	Example assertion group lines 31 to 35: Combined totals support natural word order without mistaking each, negated totals or product prices for the budget. Full input and assertion definitions: lines 1 to 167; 16 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-45	test-proactive-cross-sell.mjs	Example assertion group lines 27 to 37: The first beach-wedding result proactively shows actual bag and shoe products. Full input and assertion definitions: lines 1 to 270; 28 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]

Identifier	Script on GitHub	Input fixture and expected assertion reference	Fresh result
TC-REG-46	test-adaptive-reliability-fixes.mjs	Example assertion group lines 40 to 50: Store navigation supplies a verified product link and honest prototype/bag steps. Full input and assertion definitions: lines 1 to 236; 28 named groups indexed in E10.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-47	test-image-delivery.ts	Current source snapshot: image delivery/fallback and catalogue product image assertions.	Pass — 24 Oct offline run. [24 Oct evidence]
TC-REG-48	test-product-images.ts	Current source snapshot: image delivery/fallback and catalogue product image assertions.	Pass — 24 Oct offline run. [24 Oct evidence]

Concrete automated cases

These examples expose the inputs and expected outputs directly. They were executed within the named fresh offline suites; their Pass status does not imply live-model or browser acceptance (see TABLE 17).

TABLE 17 Concrete automated test examples

Case	Exact controlled input	Expected output verified	Executable definition
AUTO1 TC-REG-12	Message asks for a formal dress and shoes under A$400 total. Fixture D250 costs A$250; S150 costs A$150.	matches; product IDs D250 and S150; total 400; remaining 0; no cart or selected outfit mutation.	test-outfit-budget.mjs; check Exact customer request returns a complete formal set at the inclusive A$400 boundary.
AUTO2 TC-REG-12	Same A$400 brief; D300 costs A$300 and S200 costs A$200.	no_match and empty products; no A$500 combination presented as affordable.	test-outfit-budget.mjs; Individually affordable items whose sum exceeds the total are not recommended.
AUTO3 TC-REG-12	Dress A$249.99 plus shoe A$150.01, then raise shoe to A$150.02 against A$400 total.	First total is exactly 400; second produces no_match.	test-outfit-budget.mjs; Currency arithmetic is exact at the cent boundary.
AUTO4 TC-REG-14	Catalogue style Elegant and Navy; input style " elegant " and colours [" navy ","NAVY"].	Canonical Elegant and ["Navy"], no issues; input object unchanged.	test-preference-validation.ts explicit input and canonical assertions.

Case	Exact controlled input	Expected output verified	Executable definition
AUTO5 TC-REG-14	Independently validate size 99; excluded Ultraviolet; Shoes with Dresses; minimum 200 and maximum 100.	needs_clarification true for every invalid hard-value or relationship case.	test-preference-validation.ts invalid preferences loop.
AUTO6 TC-REG-15	Contract bodies use limit 6, limit 7, limit 2.5; omit limit; message length 2001; padded text.	6 accepted; 7 and 2.5 rejected; omitted limit defaults to 6; length 2001 rejected; padding trimmed.	test-recommendation-contract.ts case array and default/trim assertions.
AUTO7 TC-REG-16	P1002 at A$55 with recorded Lavender; supply a FAKE product ID or product_type as unsupported evidence reason.	Valid grounded name/colour rendered; forged product and unsupported reason rejected.	test-grounded-recommendation.ts explicit evidence and assert.throws checks.
AUTO8 TC-REG-46	Known TOP Silver Blouse at A$170; ask how to select size 12/save it and whether checkout exists.	answered; verified /products/TOP and /cart sources; Add to bag guidance; checkout boundary; no outfit mutation.	test-adaptive-reliability-fixes.mjs Store navigation assertion group.
AUTO9 TC-REG-46	Gift for Dad with unknown size, blue preference, no jewellery; controlled plan incorrectly treats context as blockers.	Useful scarf result; no jewellery; unknown size retained; physical requirements remain subject to verification.	test-adaptive-reliability-fixes.mjs recipient context and physical requirement groups.
AUTO10 TC-REG-46	Customer wants regular-fit top, size 12 under A$180, to wear with owned trousers; controlled plan wrongly searches Bottoms.	Search becomes Tops; size 12 and A$180 retained; returned items are Tops.	test-adaptive-reliability-fixes.mjs A requested top overrides the model confusing owned trousers with the purchase.

Reproducible customer scenarios

Each scenario starts with its specified fresh session and isolated synthetic catalogue. The harness sends the listed messages in order through the real interpreter plus shortcutPlan, finishConversationTurn and acceptReply. It records each repetition and turn. This is a model/domain integration test; it does not load the browser or query production stock.

Current assessed scenario outcome: each of the 38 scenarios has three latest passing repetitions. The final full run recorded 108 Pass, 0 Fail and 6 Blocked due to provider connection/timeouts. S3-B, S3-C and S4-A were then each repeated three times, all Pass. Earlier 17 October failures and the first 24 October attempt are retained in the evidence. This is not a claim of one uninterrupted 114/114 run. [E15] (see TABLE 18).

TABLE 18 Baseline customer scenario specifications and results

Case	Fixture and exact message sequence	Expected outcome	Recorded results
S1-A Complete brief, cheaper, material	Default catalogue in scripts/fixtures/customer-scenarios.mjs. 1. Show me black dresses in size 10 under $150. I only need a dress, no other items. 2. Show me cheaper options than the first dress, keeping my other requirements. 3. What material is the first dress in the new results?	Search immediately; keep Black, Dresses, size 10 and A$150. Cheaper options cost less than the first result. Answer material about the new first item; no extras or cart mutation.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
S1-B Change colour and price only	Default catalogue in scripts/fixtures/customer-scenarios.mjs. 1. Show me black dresses in size 10 under $150. 2. Actually, blue instead of black. Keep everything else. 3. My maximum is now $100.	Replace Black with Blue, then change only maximum price to A$100. Retain Dresses and size 10; do not add a second colour or infer a size.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
S1-C Deliberate no match	Default catalogue in scripts/fixtures/customer-scenarios.mjs. 1. Show me black dresses in size 10 under $1. 2. Keep the $1 limit. Don't show other colours or sizes.	No eligible result at A$1. Keep Black and size 10 after the follow-up; return no over-budget or wrong-size cards.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
S2-A Guided beach wedding shopping	Default catalogue in scripts/fixtures/customer-scenarios.mjs. 1. I need a dress for a beach wedding under $150, but I don't know which colour or style. 2. Let's try blue. I don't know my size yet; can I browse first? 3. Compare the first and second dresses and explain the differences. 4. I'd prefer something more relaxed. Keep blue and the same budget.	Allow browsing before size is known. Retain beach wedding and A$150, then Blue. Compare the two actual displayed dresses on recorded facts; retain the brief when relaxing style.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]

Case	Fixture and exact message sequence	Expected outcome	Recorded results
S2-B Reject a dress across searches	Default catalogue in scripts/fixtures/customer-scenarios.mjs. 1. Show me dresses for work under $120. 2. I don't like the first dress. Don't show it again. 3. Show me other options, still for work under $120.	Retain the rejected first product ID and exclude it from the later search; keep Work and A$120. Do not reject all dresses.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
S2-C Zip question becomes explicit must-have	Default catalogue in scripts/fixtures/customer-scenarios.mjs. 1. Show me bags under $100. 2. Does the first bag have a zip closure? 3. I only want bags with a zip closure. Keep the $100 limit.	First answer closure from specifications without changing requirements; only the third message makes zip closure mandatory. Unknown is not true.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
S3-A Uncertain customer to similar dresses	Default catalogue in scripts/fixtures/customer-scenarios.mjs. 1. I feel like buying something new, but I have no idea what I want. 2. Something for everyday wear. Maybe a dress, but I'm open to ideas. 3. Let's start with dresses. Any colour, and I haven't set a budget. 4. I like the first dress. Show me similar options under $100.	Help discovery without guessing category, size or budget. Once Dresses is chosen, show options. Liking an item does not own/add it. Retain reference and A$100 during similar search.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
S3-B Browse without preferences	Default catalogue in scripts/fixtures/customer-scenarios.mjs. 1. I'd like to browse dresses for inspiration. 2. I'm not sure about colour, size or budget. Just show me some options.	Show a manageable assortment without forcing colour, size or budget. Assess variety against fixture possibilities separately from factual correctness.	Pass — three latest real-model/fixture repetitions. Provider timeouts in the full run were followed by three passing fresh repetitions; original failures retained. [24 Oct evidence]
S3-C Switch from dresses to bags	Default catalogue in scripts/fixtures/customer-scenarios.mjs. 1. Show me blue dresses in size 10 under $150. 2. Actually, forget dresses. I want a black bag under $80.	Start a new Black bag search under A$80. Clear dress size 10, Blue and dress category; do not add a dress to the outfit.	Pass — three latest real-model/fixture repetitions. Provider timeouts in the full run were followed by three passing fresh repetitions; original failures retained. [24 Oct evidence]

Case	Fixture and exact message sequence	Expected outcome	Recorded results
S4-A Pair a bag with an owned catalogue dress	Default catalogue in scripts/fixtures/customer-scenarios.mjs. 1. Show me dresses under $150. 2. I already own the first dress. Keep it as an owned piece in my outfit. 3. Find a bag to go with my owned dress, under $80. I don't need another dress. 4. Why does the first bag go with my dress?	Keep the exact owned dress ID, do not add it to cart, and search Bags under A$80. Explain the first bag against the owned anchor using supported facts.	Pass — three latest real-model/fixture repetitions. Provider timeouts in the full run were followed by three passing fresh repetitions; original failures retained. [24 Oct evidence]
S4-B Style an external owned garment	Default catalogue in scripts/fixtures/customer-scenarios.mjs. 1. I already own a plain navy midi dress from another shop. I want shoes to match it for a wedding, under $100. 2. I don't have a product link or photo. Please work from that description.	Preserve the customer description where supported and acknowledge limitations. No invented product ID, material or exact shade. Record a feature gap if external-garment state cannot support the intended journey.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
S4-C Pair shoes with two owned pieces	Default catalogue in scripts/fixtures/customer-scenarios.mjs. Owned dress setup messages are included below. 1. Show me dresses under $150. 2. I already own the first dress. Keep it as an owned piece in my outfit. 3. Show me bags. 4. I already own the first bag too. Keep it as an owned piece in my outfit. 5. Find shoes to go with both my owned dress and bag, under $120. 6. No extra recommendations after the shoes.	Retain both owned product IDs, search shoes under A$120, and stop extras when asked. Do not replace owned items or charge their prices as new purchases.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
S5-A Pockets justify A$25 more	Fixture fixtures["S5-A"] in that file; first row is focused except S5-B. 1. Does this dress have pockets? 2. Compare a replacement with pockets under $150. 3. What does the extra cost give me? 4. Pockets would be useful.	VA-ORIGINAL costs A$95 without pockets; VA-POCKETS costs A$120 with pockets. Explain the verified gain and A$25 extra; no generic quality inference or cart mutation.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
S5-B Replacement preserves all hard requirements	Fixture fixtures["S5-B"] in that file; first row is focused except S5-B. 1. I need a blue dress for a beach wedding in size 10, under $150. 2. Compare a replacement for VB-ORIGINAL Dress with pockets. Keep the beach wedding occasion, blue, size 10 and the $150 maximum as requirements. 3. Would this still meet my beach wedding requirements?	Only VB-ELIGIBLE may qualify as the pocketed replacement: Blue, Beach Wedding, size 10, A$140. Exclude wrong occasion, colour, size and A$160 decoys; resolve the named original.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]

Case	Fixture and exact message sequence	Expected outcome	Recorded results
S5-C No demonstrated improvement	Fixture fixtures["S5-C"] in that file; first row is focused except S5-B. 1. Can you find a better replacement for this bag under $100? 2. A secure zip closure matters most. Does spending more improve that? 3. If there is no useful improvement, I'll keep this one.	VC-ORIGINAL A$60 and VC-SAME A$90 have the same recorded closure/features. Do not claim a useful improvement from price alone. Respect keeping the original.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
S5-D Same-price and cheaper adjustable alternatives	Fixture fixtures["S5-D"] in that file; first row is focused except S5-B. 1. Find a replacement for this bag with an adjustable strap, at or below $80. 2. Which options improve adjustability without costing more? 3. Explain the price difference for the cheaper one.	VD-EQUAL A$80 and VD-CHEAPER A$65 improve adjustable strap over VD-ORIGINAL A$80. Exclude VD-OVER A$95; explain A$0 extra and A$15 less accurately.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
S5-E Lining trade-off then both required	Fixture fixtures["S5-E"] in that file; first row is focused except S5-B. 1. Find a replacement with pockets under $150. 2. What would I gain and what would I lose with that replacement? 3. I need lining too. Keep both lining and pockets as must-haves, still under $150.	VE-TRADEOFF A$120 gains pockets but loses lining versus VE-ORIGINAL A$95. Explain A$25 increase; after both become must-haves return honest no-match.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
S5-F Unknown lining cannot prove improvement	Fixture fixtures["S5-F"] in that file; first row is focused except S5-B. 1. Is this dress lined? 2. Find a replacement under $150 that adds lining compared with this dress. 3. Can you confirm that the original has no lining?	VF-UNKNOWN has lining null, not false. A lined candidate does not prove an improvement over unknown. Keep unknown explicitly and identify synthetic provenance.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
S6-A Shoe size and construction details	Default catalogue in scripts/fixtures/customer-scenarios.mjs. 1. Show me black shoes for work in size 38 under $120. I only need shoes. 2. What are the heel height and sole material of the first pair? 3. Show me a cheaper option than the first pair, keeping black, work, size 38 and the same maximum.	Use recorded shoe size 38, Black, Work and A$120. Answer heel height 4 cm and sole Rubber from the referenced fixture; cheaper options retain constraints and no extras.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]

Case	Fixture and exact message sequence	Expected outcome	Recorded results
S6-B Silver necklaces and fastening facts	Default catalogue in scripts/fixtures/customer-scenarios.mjs. 1. Show me silver necklaces under $80. I only need a necklace. 2. What material and fastening does the first necklace have? 3. Show me other necklaces, still silver and under $80.	Return Silver necklaces under A$80 without clothing-size questions. Answer Silver-tone Metal and clasp from the exact first necklace; do not claim sterling silver.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]

Extended scenarios in the current harness

The suite contains 20 baseline scenarios plus 18 extensions (five budget, six category-default, four collection and three matching-set cases). Initial B5 expected an obsolete optional style question; it was corrected to the implemented grouped-discovery contract. Matching-set wording and budget-only refinement failures were repaired and retested. E15 preserves the initial failures and subsequent outcomes (see TABLE 19).

TABLE 19 Extended customer scenario specifications and results

Case	Fixture and exact messages	Expected output	Execution status
B1	fixtures["B1"] 1. I want a dress and shoes for formal events, my budget for both of them is 400 aud 2. Make the combined budget $350 instead. Keep everything else. 3. Make the combined budget 400 aud again, keeping everything else. 4. What material is the dress in this set?	BUDGET-DRESS A$250 plus BUDGET-SHOES A$150 totals A$400. A$350 yields no-match; restoration to A$400 recovers the set. Answer material about BUDGET-DRESS without a new search.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
B2	fixtures["B2"] 1. Find a dress and shoes for formal events under $400 total.	OVER-DRESS A$300 plus OVER-SHOES A$200 exceeds A$400; return no-match even though each item is individually under the total.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
B3	fixtures["B3"] 1. Find a black dress in size 10 and silver shoes in size 38 for formal events under $400 total. 2. Show the same set with a combined budget of $399. Keep the colours and sizes.	COLOUR-DRESS Black size 10 A$250 plus COLOUR-SHOES Silver size 38 A$150 qualifies at A$400. Keep separate sizes/colours and return no-match at A$399.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]

Case	Fixture and exact messages	Expected output	Execution status
B4	fixtures["B4"] 1. Build a complete outfit with a dress, shoes and bag for formal events under $350 total.	THREE-DRESS A$200, THREE-SHOES A$100 and THREE-BAG A$50 form all three requested roles at exactly A$350.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
B5	fixtures["B5"] 1. I want a dress and shoes for formal events, my budget for both of them is 400 aud 2. I would like blue dresses. Keep the shoes as they are and the 400 AUD total. 3. zibble flomp splatch 4. Only show flat shoes. Keep the blue dress options and the $400 total. 5. What material is DISC-WRAP Dress from earlier?	Retain the A$400 grouped dress/shoe request. Refine dress to Blue and shoes to Flats independently. Gibberish must not change state/context/history. Answer earlier DISC-WRAP material correctly.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
CAT-A	categoryDefaultsCatalogue and source options/initialState 1. I want a dress for work under 200 aud 2. Under 120 aud, please.	Use the implementation category default Women/Dresses; retain Work and refine A$200 to A$120 without asking a collection checkpoint.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
CAT-B	categoryDefaultsCatalogue and source options/initialState 1. I want a skirt for work under 200 aud	Resolve skirt to Clothing/Bottoms with skirt product type, not an invented Skirts hierarchy; category default is Women.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
CAT-C	categoryDefaultsCatalogue and source options/initialState 1. Men's, please. 2. I want a dress for work under 200 aud	Begin with the legacy pending collection checkpoint and complete dress brief defined by the source initialState. Men response gives no-match; a new complete dress request uses category default Women while preserving explicit collection history.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]

Case	Fixture and exact messages	Expected output	Execution status
CAT-D	categoryDefaultsCatalogue and source options/initialState 1. I want men's dresses for work under 200 aud 2. Show more dresses. 3. I want a dress for work under 200 aud	Explicit Men remains for the first two dress requests, with honest no-match. A fresh complete unqualified dress brief uses the implemented Women category default.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
CAT-E	categoryDefaultsCatalogue and source options/initialState 1. I want a dress shirt for work under 200 aud	Dress shirt remains a shirt request with collection discovery; do not convert it to Dresses.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
CAT-F	categoryDefaultsCatalogue and source options/initialState 1. Show men's clothing for all recommendations, for work under 200 aud 2. I want a dress for work under 200 aud	A shared Men restriction persists. An implicit Women dress default must ask for clarification rather than override the shared restriction.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
COL-A	collectionCatalogue; freshCollection true. 1. I dont know what I want, can you show me some formal clothes 2. Men's, please.	First reply asks collection without invented size/budget. Then use Men and Formal. For D retain A$100 and exclude Red.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
COL-B	collectionCatalogue; freshCollection true. 1. I dont know what I want, can you show me some formal clothes 2. Women's, please.	First reply asks collection without invented size/budget. Then use Women and Formal. For D retain A$100 and exclude Red.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]

Case	Fixture and exact messages	Expected output	Execution status
COL-C	collectionCatalogue; freshCollection true. 1. I dont know what I want, can you show me some formal clothes 2. Both collections, please.	First reply asks collection without invented size/budget. Then use All and Formal. For D retain A$100 and exclude Red.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
COL-D	collectionCatalogue; freshCollection true. 1. I dont know what I want, can you show me some formal clothes 2. Women's, under $100, no red	First reply asks collection without invented size/budget. Then use Women and Formal. For D retain A$100 and exclude Red.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
SET-A	matchingSetCatalogue and matchingSetState; SET-JACKET A$800 is focused. 1. can you find me bottoms to match the jacket, or some shirt as well, total budget is 500 aud 2. Make the combined budget 600 aud, keep the same pieces and jacket.	Two requested groups: SET-BOTTOM A$240 and SET-SHIRT A$260. Keep SET-JACKET as anchor, not a purchase; totals fit A$500 then A$600. No outfit mutation.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
SET-B	matchingSetCatalogue and matchingSetState; SET-JACKET A$800 is focused. 1. Please find some bottoms and a shirt that go with this jacket, total budget 500 aud. 2. Make the combined budget 600 aud, keep the same pieces and jacket.	Two requested groups: SET-BOTTOM A$240 and SET-SHIRT A$260. Keep SET-JACKET as anchor, not a purchase; totals fit A$500 then A$600. No outfit mutation.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]
SET-C	matchingSetCatalogue and matchingSetState; SET-JACKET A$800 is focused. 1. Can I get some bottoms, and also a shirt to match the jacket? My total budget is 500 AUD. 2. Make the combined budget 600 aud, keep the same pieces and jacket.	Two requested groups: SET-BOTTOM A$240 and SET-SHIRT A$260. Keep SET-JACKET as anchor, not a purchase; totals fit A$500 then A$600. No outfit mutation.	Pass — three latest real-model/fixture repetitions. See E15 for all attempts and turn-level assertions. [24 Oct evidence]

New AI feature coverage

The regression and browser cases cover suggested replies, optional proactive extras, pause/resume, anchor selection, combined budgets, saved-look review, fit guidance, product explanations and value alternatives. These implement UC01–UC06 paths; cart actions are UC07. Current browser results follow below and retain any failed or partially established expectation (see TABLE 20).

TABLE 20 AI feature coverage and linked tests

Feature and objective	Existing automated coverage	Browser cases	Behaviour and evidence boundary
Guided discovery and suggested replies — UC01/UC02	TC-REG-07, 18, 42	TC-AIX-01, 06	Focused questions, skipping uncertainty, browsing with unknown size/budget, and retaining supplied preferences. Browser checks verify visible chips and the message actually submitted.
Proactive complementary suggestions — UC04	TC-REG-45, 46	TC-AIX-02 to 04	28 named checks in each cited suite cover optional additions, references, dismissal, pause/resume, exclusions and total-budget safeguards. Client requested more proactive additions; team reports the feedback addressed. [E13]
Suggested outfit and look review — UC04/UC06	TC-REG-09 to 12	TC-AIX-05	Complete affordable pairings, selection/replacement, owned-item treatment, exact totals and review of saved pieces. A suggestion must not automatically select or purchase items.
Explanations and fit guidance — UC02/UC03/UC05	TC-REG-33, 40 to 42, 46	TC-AIX-06, 07	Catalogue-based reasons, disclosed differences, fit/stock uncertainty and clear purchase navigation. Existing targeted navigation capture is in E8; it does not execute all guidance controls.
Cross-sell and value alternatives — UC04	TC-REG-19 to 22, 24, 25; S5-A to F	TC-AIX-08	Additional items preserve relevant constraints. Upgrades require supported benefits and accurate price differences; no improvement is an acceptable outcome when evidence is absent.
Memory and combined budgets — UC06/UC04	TC-REG-01, 03, 11, 12, 44 to 46; B1 to B5	TC-AIX-04, 05; S1 to S4	Keep explicit requirements, rejected items, owned/example roles and inclusive shared totals across follow-ups. The saved look and bag remain separate.

Browser execution prerequisites

Use the frozen candidate and a saved live-catalogue snapshot at 1440 by 900 and 390 by 844. Record browser/version, product IDs and actual messages. Start each case with New chat and empty look/bag unless stated otherwise; clear stored offer preferences for an independent case. Before offer tests, confirm eligible primary products, complementary stock and rules. If a prerequisite is absent, mark the dependent step Blocked with the reason. Do not change production catalogue data to manufacture a pass.

AI suggestion and guidance browser cases

The following are actual deployed-browser executions on 24 October, with saved accessibility text and screenshots in E16. Pass applies only to the actions and expectations recorded. The fixture prerequisites and model responses are retained; an intermittent failure is not erased by a later passing request (see TABLE 21).

TABLE 21 AI suggestion and guidance browser test cases

Identifier	Exact actions and input	Expected observable output	Actual result and status
TC-AIX-01 Suggested replies	New chat; choose Women if asked. Send: I do not know what I am looking for. Record visible suggested-reply labels and associated messages; click the first reply once. While processing, attempt a second click.	A focused question and up to three relevant choices appear when required by the discovery stage. The chosen message is submitted once; busy controls prevent a duplicate request. No size, colour or budget is invented. Subsequent browsing can proceed without optional answers.	Pass — discovery displayed Men/Women/Both; Women submitted once, busy choices disabled, then Everyday wear/An occasion/Show ideas. No unstated size/colour/budget was invented. [24 Oct evidence]
TC-AIX-02 Proactive products	With eligible dress and complementary catalogue records/rules, send: Show me women's dresses for a beach wedding under 200 AUD. Let me browse without choosing a size. Inspect Optional finishing touches before requesting an extra.	Eligible optional product cards appear with the correct dress anchor and recorded facts. They are labelled as a styling example until selected. The A$200 item cap is preserved; extras are priced separately unless a combined total was supplied. Look/bag stay unchanged.	Pass — primary P1178 A$95 and P1162 A$105; optional P1250 A$100 anchored to P1178, labelled styling example and priced separately. Look remained zero. [24 Oct evidence]
TC-AIX-03 Hide pause and resume	From TC-AIX-02 click Hide these suggestions, then ask: Show other dresses with the same requirements. In an independent repeat click No more extras; repeat the request and reload. Send: Resume extra suggestions. Then repeat the eligible dress request.	Hide removes the current task offer; No more extras suppresses further proactive cards and persists across reload. Explicit resumption permits a later eligible offer. Main results, constraints and look/bag are unchanged by hiding or pausing.	Pass — Hide removed current extras; No more extras persisted across reload/repeated eligible request; explicit Resume extra suggestions restored a later eligible offer. [24 Oct evidence]

Identifier	Exact actions and input	Expected observable output	Actual result and status
TC-AIX-04 Anchor and budget	From an eligible offer with two primary dresses, use Match a different dress and select the second displayed product. Record IDs and prices before/after. Independently send: Find a women's dress and a bag for a beach wedding under 200 AUD altogether.	The anchor selector refreshes extras for the chosen displayed ID without changing primary order or selecting it. For the combined request, every proposed complete purchase fits A$200, or the response honestly reports no match. Zero remaining allowance permits no additional offer.	Mixed — anchor selector changed P1178 to P1162 and refreshed extras to P1097 without changing primary order/selection. Combined A$200 request once failed with false saved-occasion conflict; later fresh local and deployed HTTP probes returned complete A$155–195 combinations. Intermittent failure retained (DF6). [24 Oct evidence]
TC-AIX-05 Pairing and saved look	Send: Find a women's dress and shoes for formal events under 400 AUD total. With a complete eligible set, inspect Suggested pairing labels; Save to look one product from each group. Ask: What do you think of my selected outfit? Remove one item through View look.	Suggested pairing covers both roles within A$400 without automatic selection. Saved purchases total correctly; owned items have no new purchase cost. Review refers to saved pieces with supported reasons. Removal updates the look and total. Bag stays unchanged.	Pass — P1284 A$240 + P1138 A$145 totalled A$385, A$15 remaining. Suggested set did not auto-select; saved-look review named actual pieces and limitations. Removing shoes left A$240/A$160. No Add to cart action. [24 Oct evidence]
TC-AIX-06 Fit and discovery guidance	Send: Show me women's clothing under 150 AUD and keep my size unknown. Then: Explain the fit labels while I browse. Does Regular guarantee that it will fit me?	Guidance explains recorded cut/fit labels without guaranteeing personal fit, stretch or measurements. Size remains unknown and A$150 stays saved. No invented feature restriction or irrelevant product-reference loop prevents continued browsing.	Pass — fit guidance explained cut labels and disclaimed personal fit/stretch/adjustability. Size stayed unknown; existing cards stayed within A$150. No product-reference loop blocked browsing. [24 Oct evidence]

Identifier	Exact actions and input	Expected observable output	Actual result and status
TC-AIX-07 Reasons and next steps	After a valid product result, ask: Why is the first item suitable for my occasion? Then: Give me its product link and explain where I choose a listed size and save it to my bag. Is checkout available? No other products.	Reasons match saved catalogue attributes and disclose mismatches. Links identify the referenced product and bag/help pages. Guidance uses Add to bag and states checkout, payment and order placement are unavailable. No automatic bag change, guaranteed size stock or extra products.	Pass — explanation for P1254 used recorded Brunch/Relaxed/Summer/A$120 facts. Product/bag/help links and listed-size/quantity/Add to bag instructions were provided. Checkout, payment, order placement and individual-size stock guarantees were explicitly unavailable; bag unchanged. [24 Oct evidence]
TC-AIX-08 Additional items and value	Choose an actual displayed product; record its ID, price and constraints. Send: Find an extra for [recorded product name]. Then independently: Compare value alternatives to [recorded product name], keeping my requirements. Record exact substituted messages.	Additional products have supported pairing reasons and preserve applicable restrictions. Value alternatives disclose verified gains, compromises and price differences. If no improvement is evidenced, state that clearly. Neither request automatically selects or purchases a product.	Fail — extra request correctly clarified category and returned P1201 A$150 with disclosed pairing mismatches. The subsequent exact value-alternatives prompt was misread as an occasion and displayed A$200/A$240 products despite the saved A$150 maximum, without the required value comparison (DF8). [24 Oct evidence]

Additional integration and acceptance cases

The HTTP and isolated dependency failures were executed on 24 October. TC-UI-01 shares the revised fixed-colour scope of TC-PD-02. TC-ACC-01 records the supplied client decision, separately from technical results. There is no assertion of full platform, administrator or deployment-source verification (see TABLE 22).

TABLE 22 Integration deployment and acceptance test cases

Identifier	Input and execution method	Expected observable output	Actual result and status
TC-HTTP-01	POST /api/conversation-search with Content-Type text/plain and body {}.	HTTP 415; controlled Send JSON error; no product retrieval or sensitive content.	Pass — local and deployed endpoints returned the specified controlled HTTP status/body. [24 Oct evidence]
TC-HTTP-02	Same route with application/json and exact malformed body {"turn":	HTTP 400 with Invalid JSON error; no crash or secrets.	Pass — local and deployed endpoints returned the specified controlled HTTP status/body. [24 Oct evidence]
TC-HTTP-03	Same route with application/json and body {}.	HTTP 400 with Invalid conversation request error; no fabricated recommendations.	Pass — local and deployed endpoints returned the specified controlled HTTP status/body. [24 Oct evidence]
TC-HTTP-04	Same route with Origin https://other.example and a local Host header; use a safe otherwise valid test request.	HTTP 403 Origin not allowed; no model/catalogue execution.	Pass — local and deployed endpoints returned the specified controlled HTTP status/body. [24 Oct evidence]
TC-HTTP-05	Send an application/json request larger than 64000 bytes. Record exact byte length and safe body.	HTTP 413 Request is too long. Boundary values 64000 and 64001 need a valid envelope to isolate the size check; invalid bodies must not be counted as successful valid-boundary requests.	Pass — local and deployed endpoints returned the specified controlled HTTP status/body. Valid request at exactly 64000 bytes returned 200; 64001 returned 413. [24 Oct evidence]

Identifier	Input and execution method	Expected observable output	Actual result and status
TC-INT-01 Related TC-AI-15	On isolated server POST a valid new conversation-search envelope for a blue bag under A$80 for a wedding; create the turn using makeTurn/newSession. Inject an OpenAI request failure during interpretation. Record injection method and request ID.	HTTP 502; code CONVERSATION_FAILED; error states the request could not be completed safely and no cart changes were made. No product claims, raw provider error or secret. Verify preserved client state.	Pass — injected model failure returned 502 CONVERSATION_FAILED, safe message and unchanged request state. [24 Oct evidence]
TC-INT-02 Related TC-AI-16	On isolated server use a valid conversation-search envelope and inject a catalogue query/connection failure after interpretation. Record the chosen test adapter and restore it afterwards; do not disrupt shared Supabase.	HTTP 503; code CATALOGUE_UNAVAILABLE; controlled safety message with no products or raw connection details. A new request can recover after the isolated catalogue service is restored.	Pass — injected catalogue failure returned 503 CATALOGUE_UNAVAILABLE; restored service returned 200 with catalogue-backed products. [24 Oct evidence]
TC-UI-01 TC-PD-02	Revised scope: compare P1031 Gold colour label/swatch, product page and cart row; verify only supported catalogue information. The product has no alternate colour variant.	Recorded single colour is consistent across product and cart; no variant-switch functionality is claimed.	Pass — fixed-colour display verified. Original multicolour interaction is superseded by this explicit scope revision. [24 Oct evidence]
TC-UI-02 TC-PD family	Execute PD1 add/remove/quantity/navigation on desktop 1440 by 900 and mobile 390 by 844. Reload the bag after saving the selected size. Record browser/device.	No cut-off or overlapping controls; counts, item subtotal and disclosed totals remain consistent; saved product/size/quantity survive supported persistence.	Pass — desktop/mobile add, separate-size rows, repeat add, increase/decrease, removal, navigation and reload persistence completed with P1031 A$90. Empty cart restored. [24 Oct evidence]
TC-INT-03	Save current live catalogue snapshot; compare every returned product ID, price, size label and availability field used in the chosen cases.	Returned facts match the snapshot; missing values remain unknown. Report aggregate stock separately from per-size stock. Product count is an observation, not a fixed expectation of 316.	Pass — deployed product IDs, names, prices and colours matched the saved current catalogue snapshot. Listed sizes do not assert per-size stock. [24 Oct evidence]

Identifier	Input and execution method	Expected observable output	Actual result and status
TC-DEP-01	Open https://abcandco.vercel.app/ from a fresh browser session. Record final URL, deployment/build identity and timestamp; compare to the frozen candidate. At 1440 by 900 and 390 by 844 open the catalogue and AI Stylist, then send: Show me a blue bag under 80 aud for a wedding. Inspect network and console. [E12]	Root page returns HTTP 200; navigation and critical assets load without errors that prevent the journey. Conversation responds through /api/conversation-search with catalogue-grounded cards or an honest no-match result. No unhandled console exception or failed critical request. If the candidate cannot be identified, mark build identity Blocked and retain observed behaviour separately.	Partial — root HTTP200; catalogue/stylist and exact blue-bag A$80 wedding request worked on desktop/mobile, returning an honest no-match. Browser log captures retained. Deployed source revision was not exposed; the local fixes are not proven deployed. [24 Oct evidence]
TC-ACC-01	21 October 2026 client meeting with Peyton Bernhardt and all team members. Salvi supplied meeting minutes, the product acceptance decision and the update that feedback was addressed. Meeting URL/build and individual demonstrated steps were not supplied. [E13]	Record the client decision, feedback, team response and demonstration scope. Keep individual technical case results and prototype limitations separately traceable.	Accepted on 21 Oct 2026, as reported by Salvi. All features approved and product accepted as ready. Sarah completes meeting build details and links the demonstrated scope.

Execution results and defect follow up

Latest execution and evidence separation

The execution summary separates offline checks, live integration results, browser observations and client acceptance (see TABLE 23).

TABLE 23 Latest test execution summary

Evidence layer	Result and scope
Offline candidate	48/48 suites Pass on 24 Oct; typecheck, lint and production webpack build Pass. Local changes and hashes are retained in E14.
Catalogue/model/HTTP	32/32 latest check IDs Pass. Initial request-context setup failures retained in attempt-1; corrected standalone read-only catalogue setup then passed.
Dependency failures	3/3 isolated HTTP checks Pass: provider 502, catalogue 503 and recovery 200. Shared credentials/services were unchanged.
Real-model customer scenarios	Full final run 108 Pass/0 Fail/6 provider-timeout Blocked; affected three scenarios repeated 3 times, 9/9 Pass. All 38 scenarios have three latest passing repetitions; all attempts retained.
Deployed browser	Product/cart and recorded guidance/offer controls passed. AIX04 has an intermittent false conflict, BR03 failed (fixed locally) and AIX08 value comparison failed to preserve the saved budget. See DF6–DF8.
Client decision	Accepted 21 October 2026, reported by Salvi; final D4 records 6:00 pm. Sarah remains responsible for independent technical review.
Coverage boundary	Eight customer use cases mapped. Full SRS certification is not claimed: administrator FR-34–40, native-browser matrix, sustained NFR thresholds and several mutation/error alternatives lack executed evidence.

Defect and retest disposition

The defect register retains the original problems, retest evidence and current disposition (see TABLE 24).

TABLE 24 Defects and retest outcomes

ID	Observed problem	Current evidence and next action
DF1	Historical dependency/setup issue	Resolved for current offline execution: 48/48 Pass. First live-catalogue harness used a request-bound helper outside HTTP; corrected setup passed. First attempt retained.
DF2	Historical hard-constraint and refinement failures	Current 38 scenario suite has three latest passing repetitions each; six provider timeouts were rerun successfully. Does not establish browser reliability for every free-text phrasing.
DF3	Matching-set wording and budget-only refinement	Local targeted fix verified by named regression assertions, 12/12 targeted model runs and the complete scenario run/recovery. Source patch retained; deployed source equality is unverified.

ID	Observed problem	Current evidence and next action
DF4	Service failures and old multicolour test	AI/database/recovery checks Pass. Multicolour requirement explicitly replaced with recorded single-colour display, aligned to UC05; no variant feature was added.
DF5	Provisional SRS/use-case mapping	Reconciled against final D4 Figure 7, pp. 62–72 and FR-01–FR-40. Eight customer cases mapped; administrator and other unproved alternatives are disclosed in the coverage limits.
DF6	Intermittent false occasion conflict on deployed combined request	Retained failure: dress+bag/beach-wedding/A$200 request reported conflicting saved occasion on a new chat. Fresh local and deployed HTTP probes later returned valid A$155–195 combinations. Cause/consistent deployed repair not established; treat as known intermittent limitation.
DF7	Browse all products did not clear deployed search state	Fail on deployed BR03: URL changed but no-results state remained. SearchExplorer now resets query and reloads through the controller. Local browser retest displayed 316 products; quality checks passed. This local fix is not asserted deployed.
DF8	Value alternatives lost saved maximum budget	Fail on deployed AIX08: exact value-comparison request was interpreted as occasion text and cards at A$200/A$240 appeared after the saved A$150 maximum. Budget/intent regression requires a fix and deployed confirmation; current technical acceptance criterion is not met.

Existing browser evidence

The older Figure 1 records a 22 October navigation answer for historical comparison. The new E16 captures actual desktop/mobile product, cart and AI controls on 24 October. These include quantity arithmetic, persistence, optional-offer controls and saved-look review; none establishes payment or order fulfilment.

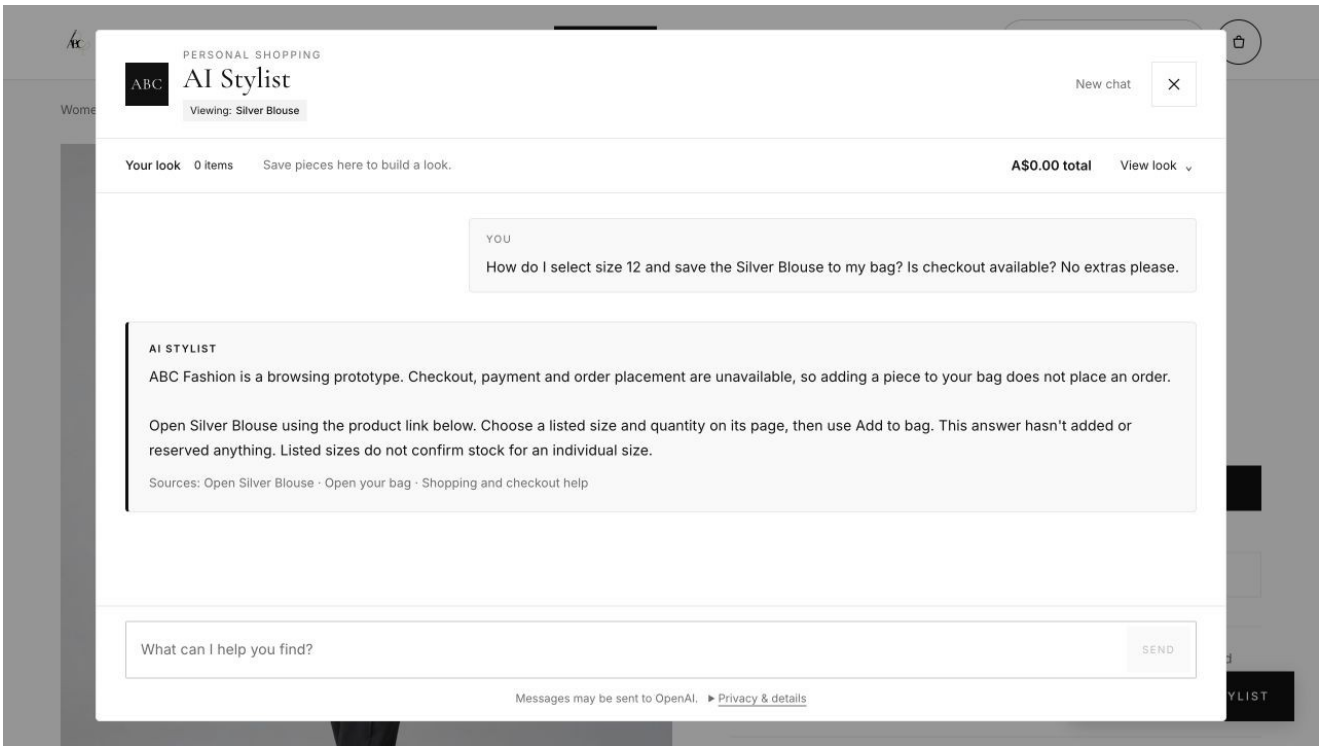


Figure 1 Existing production navigation response for Silver Blouse on 22 October 2026

Client acceptance and feedback

Peyton Bernhardt and all project team members attended the 21 October 2026 meeting. Salvi confirms that all features were approved and Peyton accepted the product as ready. The supplied minutes describe the current version as satisfactory and suggest more proactive recommendations for additional items. The team reports that it has addressed the feedback. [E13]

Sponsor meeting feedback and response to feedback

The feedback record links the client comments to team responses and available verification (see TABLE 25).

TABLE 25 Client feedback and team responses

Client feedback	Team response and verification
User instructions were important and thoughtful.	Retain the demonstrated instructions in the final product and user documentation. Sarah checks consistency with the submitted version.
Clear recommendation explanations and the options offered were particularly impressive.	Preserve catalogue-based reasons, disclosed differences and clear options. Existing TC-REG-33, 40 and 41 passed; TC-AIX-07 specifies the complete browser interaction.
The new version improved accuracy and user experience.	Retain constraint, state and budget safeguards. E2 records 46/46 offline suites passing on 22 October; E7/E8 preserve earlier problems and targeted fixes without implying a measured accuracy percentage.
The UI looked great.	Retain the approved design. Verify final desktop/mobile behaviour and new AI controls through TC-UI and TC-AIX cases; client approval is recorded separately from those observed results.
More proactive recommendations for additional items were suggested; current version satisfactory.	Addressed according to the team update. Current optional finishing touches show complementary products with anchor, dismissal, pause/resume and budget controls. TC-REG-45 and 46 passed; TC-AIX-02 to 04 specify remaining browser verification.

Acceptance record and final review

Client decision: accepted as ready on 21 October 2026, as reported by Salvi and transcribed in E13. The final D4 document records the meeting at 6:00 pm (page 55). The URL supplied for current testing is https://abcandco.vercel.app/. The exact source revision demonstrated at that meeting was not supplied. Sarah Thomas owns the final technical review; no signature or review completion is fabricated.

Handover consists of this editable report, PDF, offline evidence ZIP and the corresponding candidate/evidence snapshots. The new eight-use-case mapping and SRS coverage limits are documented in E17. Product acceptance is retained, while technical limitations remain explicit for the team and assessor.

Evidence register and handover

E0–E13 retain their fixed historical publication links. E14–E17 link to the new 24 October evidence and use case source in the same GitHub repository. Reviewers require private-repository access; the ZIP contains an offline copy for submission without GitHub authentication. Source references use publication commits; candidate manifests identify the code actually executed (see TABLE 26).

Project repository: ABC Fashion AI Assistant

TABLE 26 Evidence register and source links

ID	Evidence on GitHub and meaning
E0	Original testing specifications Original supplied 23-page Part B PDF. Historical outcomes and original IDs.
E1	COMP3850 deliverables brief Assessment brief pp. 19 to 22, 36 and 39. Plan, case, revision and formatting requirements.
E2	Fresh offline regression log Complete 46-suite run on 22 Oct 2026. Console mirror is linked with E3.
E3	Execution manifest Git HEAD, dirty tree status, source hashes before/after and execution timestamps. Regression console record
E4	Original recommendation results Historical single-message development endpoint; separate from the current customer route.
E5	Historical 41-suite regression log Saved 41/41 historical run. The later blocked log named in the original PDF is unavailable.
E6	Historical customer scenario report Historical assessment with per-run summary, saved fixture and transcripts. Scenario transcripts
E7	Adaptive browser baseline report Existing live baseline, environment, catalogue audit, verification and transcripts.
E8	Adaptive reliability fixes and retests Existing targeted retest, browser capture and probes. Iterative scope retained.
E9	Assessed scenario run outcomes Original run-level outcomes, separate from the exploratory pilot summary.
E10	Regression and scenario source index Exact definitions, line ranges and fixture sources. Script links in Part B open the same fixed revision. Original case transcription Regression scripts Scenario fixtures
E11	Customer conversation route HTTP contract and the legacy development-only boundary. Both routes linked below. Legacy recommendation route
E12	Original test plan Prior Part A. Team/client p. 5; tools/deployment pp. 3 to 4; D3 milestone p. 5.
E13	Client acceptance and meeting feedback 21 October minutes, attendance, acceptance and response supplied by Salvi. Meeting URL/build and time/format remain unspecified.
E14	24 October candidate, offline/HTTP/catalogue/model checks, controlled failures, quality checks and current source snapshots

ID	Evidence on GitHub and meaning
E15	All 24 October customer-scenario attempts and later timeout recovery; 38 cases with three latest passing repetitions
E16	24 October browser-results.jsonl plus per-case screenshots and accessibility captures
E17	Final supplied D4 document: Figure 7p. 62, UC descriptions, pp.63–72, SRS, pp.38–53; eight-case alignment and explicit coverage limits— source PDF extract

Preserve the eight-use-case mapping, revision history and evidence hyperlinks when merging this testing section into D4. The diagram/descriptions are requirement sources; their inclusion does not authorize tests, signatures or acceptance decisions that have not occurred.